The Cosine Rule

Exam code: 0607

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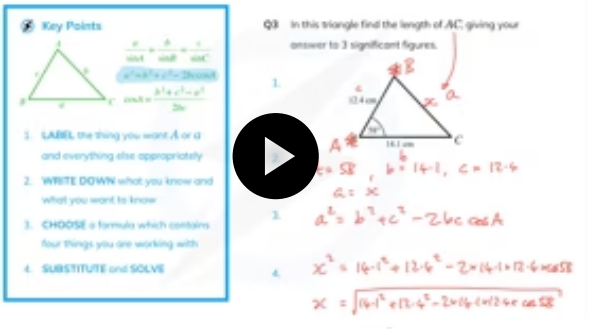

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Cosine rule

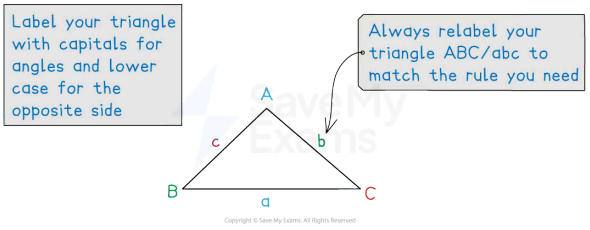
What is the cosine rule?

Help me understand

- The **cosine rule** is used in **non right-angled triangles**
  - It allows us to find missing side lengths or angles
- It states that for any triangle

$$a^2 = b^2 + c^2 - 2bc \cos A$$

- Where
  - a is the side **opposite** angle A
  - b and c are the other two sides
    - b and c are either side of angle A
    - A is the angle between them



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

How do I use the cosine rule to find a missing length?

- and you want to find the **opposite side**, *a*

- **Start by labelling your triangle** with the angles and sides
  - Angles have upper case letters
  - Sides **opposite** the angles have the equivalent lower case letter

- **Substitute** values into $a^2 = b^2 + c^2 - 2bc \cos A$
  - Make *a* the subject (don't forget to **square root**)

How do I use the cosine rule to find a missing angle?

- Use the **cosine rule** for angles
  - when you have **all three sides**
  - and you want to find an angle

- It helps to **rearrange** the formula as follows, by adding $2bc \cos A$ to both sides then making $\cos A$ the subject

$$\begin{aligned} a^2 &= b^2 + c^2 - 2bc \cos A \\ a^2 + 2bc \cos A &= b^2 + c^2 \\ 2bc \cos A &= b^2 + c^2 - a^2 \\ \cos A &= \frac{b^2 + c^2 - a^2}{2bc} \end{aligned}$$

- Use the formula $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$ to find the **unknown angle** *A*
  - Remember, *A* is the angle **between** sides *b* and c
    - (you may need to relabel the triangle)

  - You will need to use **inverse cosine** at the end, $\cos^{-1}(...)$

- Unlike the sine rule, there is no ambiguous case of the cosine rule

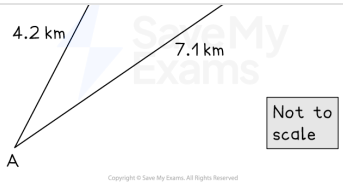
Examiner Tips and Tricks

You are given the cosine rule in the form $a^2 = b^2 + c^2 - 2bc \cos A$ on the formula sheet.

Getting an error on your calculator when finding an angle may mean you have rearranged the formula incorrectly.

Worked Example

The following diagram shows triangle ABC, where $\text{AB} = 4.2 \text{ km}$, $\text{BC} = 3.8 \text{ km}$ and $\text{AC} = 7.1 \text{ km}$.



Calculate the value of angle ABC.

Answer:

The side opposite the angle is 7.1, so $a = 7.1$

The sides 4.2 and 3.8 are b and c (in either order)

Use the cosine rule in the rearranged form

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos\theta = \frac{4.2^2 + 3.8^2 - 7.1^2}{2(4.2)(3.8)}$$

$$\theta = \cos^{-1}\left(\frac{4.2^2 + 3.8^2 - 7.1^2}{2(4.2)(3.8)}\right)$$

$$\theta = 125.04699...$$

$$\theta = 125.0° \text{ (to 1 d.p.)}$$

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